

1.CCNP Enterprise作业一——交换基础

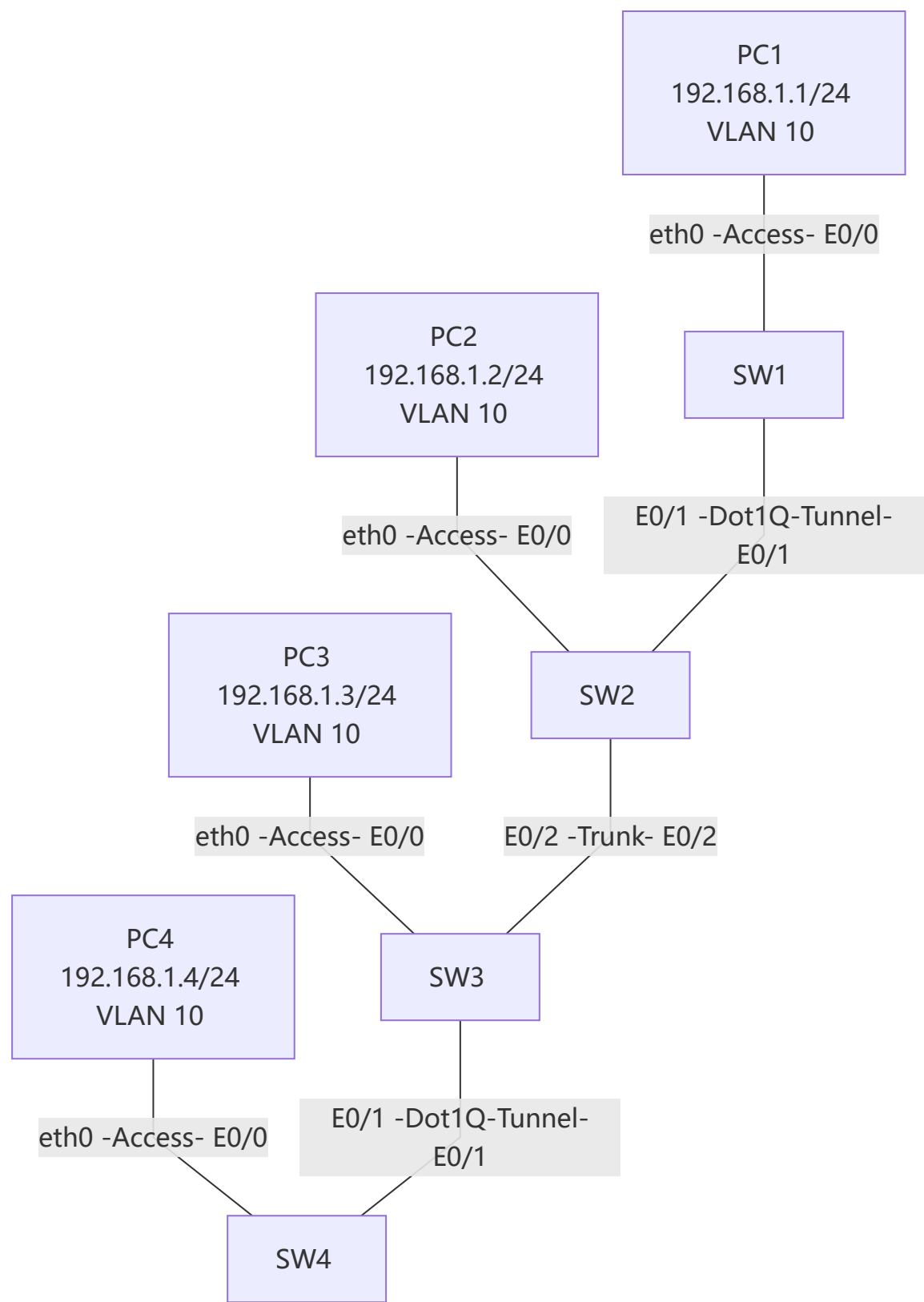
作业主题

交换基础

作业目标

- 1. VLAN 的规划与配置
 - 2. Access 端口与 Trunk 端口的配置
 - 3. 完成QinQ的配置
-

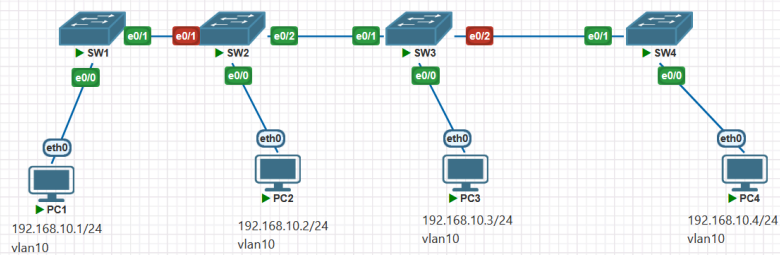
实验拓扑



需求

1. 根据实验拓扑完成设备连接与基础配置
2. 完成 PC1、PC2、PC3、PC4 的 IP 地址配置

3. 完成 VLAN 10 的创建与规划
4. 完成交换机 Access 端口配置
5. 完成交换机 Trunk 端口配置
6. 完成 Dot1Q Tunnel 端口配置
7. 完成 QinQ 配置



sw1

```
!
! Last configuration change at 03:51:31 UTC Thu Sep 10 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
```

```
!  
!  
!  
no ip domain-lookup  
ip cef  
no ipv6 cef  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface Ethernet0/0  
no shutdown  
switchport access vlan 10  
switchport mode access  
!  
interface Ethernet0/1  
no shutdown  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Ethernet0/2  
no shutdown  
!  
interface Ethernet0/3  
no shutdown  
!  
interface Ethernet1/0  
no shutdown  
!  
interface Ethernet1/1  
no shutdown  
!  
interface Ethernet1/2  
no shutdown  
!  
interface Ethernet1/3  
no shutdown
```

```
!  
interface Ethernet2/0  
no shutdown  
!  
interface Ethernet2/1  
no shutdown  
!  
interface Ethernet2/2  
no shutdown  
!  
interface Ethernet2/3  
no shutdown  
!  
interface Ethernet3/0  
no shutdown  
!  
interface Ethernet3/1  
no shutdown  
!  
interface Ethernet3/2  
no shutdown  
!  
interface Ethernet3/3  
no shutdown  
!  
ip forward-protocol nd  
!  
ip tcp synwait-time 5  
ip http server  
ip http secure-server  
!  
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr  
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr  
!  
!  
!  
!  
!  
control-plane  
!  
!  
line con 0  
exec-timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0  
exec-timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4  
login  
!  
!
```

```
!  
end
```

sw2

```
!  
! Last configuration change at 03:51:31 UTC Thu Sep 10 2026  
!  
version 15.2  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
service compress-config  
!  
hostname SW2  
!  
boot-start-marker  
boot-end-marker  
!  
!  
!  
no aaa new-model  
!  
!  
!  
!  
no ip icmp rate-limit unreachable  
!  
!  
!  
no ip domain-lookup  
ip cef  
no ipv6 cef  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
!  
!  
interface Ethernet0/0  
no shutdown  
switchport access vlan 10  
switchport mode access  
!  
interface Ethernet0/1  
no shutdown  
switchport access vlan 88  
switchport mode dot1q-tunnel  
no cdp enable  
!  
interface Ethernet0/2  
no shutdown  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Ethernet0/3  
no shutdown  
!  
interface Ethernet1/0  
no shutdown  
!  
interface Ethernet1/1  
no shutdown  
!  
interface Ethernet1/2  
no shutdown  
!  
interface Ethernet1/3  
no shutdown  
!  
interface Ethernet2/0  
no shutdown  
!  
interface Ethernet2/1  
no shutdown  
!  
interface Ethernet2/2  
no shutdown  
!  
interface Ethernet2/3  
no shutdown  
!  
interface Ethernet3/0  
no shutdown  
!  
interface Ethernet3/1  
no shutdown  
!  
interface Ethernet3/2  
no shutdown
```

```

!
interface Ethernet3/3
no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
ip http secure-server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
!
end

```

sw3

```

!
! Last configuration change at 04:09:10 UTC Thu Sep 10 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW3
!
boot-start-marker
boot-end-marker
!
!

```

```
!  
no aaa new-model  
!  
!  
!  
!  
!  
no ip icmp rate-limit unreachable  
!  
!  
!  
no ip domain-lookup  
ip cef  
no ipv6 cef  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
!  
!  
!  
!  
!  
!  
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!  
!  
!  
!  
!  
!  
!  
interface Ethernet0/0  
no shutdown  
switchport access vlan 10  
switchport mode access  
!  
interface Ethernet0/1  
no shutdown  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Ethernet0/2  
no shutdown  
switchport access vlan 88  
switchport mode dot1q-tunnel  
no cdp enable  
!  
interface Ethernet0/3  
no shutdown  
!
```

```
interface Ethernet1/0
no shutdown
!
interface Ethernet1/1
no shutdown
!
interface Ethernet1/2
no shutdown
!
interface Ethernet1/3
no shutdown
!
interface Ethernet2/0
no shutdown
!
interface Ethernet2/1
no shutdown
!
interface Ethernet2/2
no shutdown
!
interface Ethernet2/3
no shutdown
!
interface Ethernet3/0
no shutdown
!
interface Ethernet3/1
no shutdown
!
interface Ethernet3/2
no shutdown
!
interface Ethernet3/3
no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
ip http secure-server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
```

```
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
!
end
```

sw4

```
!
! Last configuration change at 03:51:33 UTC Thu Sep 10 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW4
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface Ethernet0/0  
no shutdown  
switchport access vlan 10  
switchport mode access  
!  
interface Ethernet0/1  
no shutdown  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Ethernet0/2  
no shutdown  
!  
interface Ethernet0/3  
no shutdown  
!  
interface Ethernet1/0  
no shutdown  
!  
interface Ethernet1/1  
no shutdown  
!  
interface Ethernet1/2  
no shutdown  
!  
interface Ethernet1/3  
no shutdown  
!  
interface Ethernet2/0  
no shutdown  
!  
interface Ethernet2/1  
no shutdown  
!  
interface Ethernet2/2  
no shutdown  
!  
interface Ethernet2/3
```

```

no shutdown
!
interface Ethernet3/0
no shutdown
!
interface Ethernet3/1
no shutdown
!
interface Ethernet3/2
no shutdown
!
interface Ethernet3/3
no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
ip http secure-server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
!
end

```

7. 验证 PC1、PC2、PC3、PC4 的连通性

	pc1	pc2	pc3	pc4
pc1		不通	不通	通

	pc1	pc2	pc3	pc4
pc2	不通		通	不通
pc3	不通	通		不通
pc4	通	不通	不通	

8. 分析 PC1 与 PC4 的通信结果（二层帧交付过程）

答：能通信，pc1 通过access接口进入sw1打上vlan10标签(c-tag),在进过sw e0/2接口又被打上s-tag 88标签继续在trunk链路上传输，直到从sw3的e0/2接口出的时候，被剥离s-tag vlan88标签，再从sw4的e0/0的access接口出的时候，被剥离vlan10的标签

9. 分析PC1与PC2能否进行通信，如果可以描述过程，如果不可以请讲述理由

不可以，sw2此时收到的包已经被打上了vlan88的标签，不识别内层的c-tag标签